OligoTox-Hydrocephalus — Methodology

Materials and methods for the hydrocephalus / CSF-dynamics oligonucleotide toxicity dataset
NIH/NCATS OligoTox Open Data Challenge — Phase 2 · release v1.0

1 Design and no-fabrication policy

This is a curated dataset; no new wet-lab experiment was performed. The methods below are therefore of two kinds, kept strictly separate: **(a)** the experimental and reporting methods of the source studies, recorded so a user knows how each measurement was generated, and **(b)** the curation methods used here. Conflating them would let curation choices masquerade as experimental fact.

No value in this dataset was recalled from memory or inferred. Every number, sequence, denominator, quotation and identifier was copied by a script from a payload committed to the repository, or transcribed from a full text also committed, and carries the exact locus it came from. Where a source is silent the field is NOT_REPORTED. **No number was read off a figure**: where a value is published only graphically, readout_value is NOT_REPORTED and readout_is_qualitative is TRUE.

2 Endpoint definition

Inclusion required a ventricular, CSF-volume, CSF-pressure or CSF-composition outcome for an identified oligonucleotide, or a population baseline for such an outcome. **Tier A** is the core endpoint: hydrocephalus, ventriculomegaly, ventricular volume, shunt or drain placement. **Tier B** is CSF-dynamics adjacent: raised intracranial pressure, papilloedema, aseptic or chemical meningitis, arachnoiditis, CSF protein or cell-count rise, post-lumbar-puncture syndrome. General neurological adverse events with no CSF or ventricular readout contribute nothing. **Route of administration is not itself evidence** of the endpoint.

3 Source identification and retrieval

Sources were sought across six deliberately different modalities, because each is biased differently: literature finds published positives, a registry finds protocol-collected events including zeros, a spontaneous-reporting system finds post-marketing signals, labels find regulator-adjudicated risks, an epidemiological cohort finds the disease baseline, and the INN lists give chemistry. Trial selection is a query, not a judgement: ClinicalTrials.gov was asked for **every** trial of any of 41 oligonucleotide therapeutics with posted results, yielding 155 usable trials. An earlier hand-picked list of 23 biased the dataset toward compounds already suspected of causing the endpoint and omitted most of the negative evidence. Every retrieved payload is committed, so the dataset rebuilds offline.

4 Extraction

Six of eight components are deterministic parsers rather than transcription. Each copies values from a named path in a committed payload, which becomes the row's source_location. Two components are curated from prose a human read; each of their rows stores the **verbatim sentence** it was taken from, so value and evidence travel together.

Component	Rows	Method
extract_ctgov.py	—	Walks resultsSection.adverseEventsModule; one row per trial × term × arm, copying numAffected and numAtRisk from the named JSON path. MedDRA terms are never normalised or spelling-corrected.
extract_ctgov_outcomes.py	—	Pre-specified ventricular MRI outcome measures — the only rows where the ventricles were measured rather than incidentally observed. Selection is an explicit allow-list; a pattern match would sweep in CSF pharmacokinetics and neurofilament.

extract_faers.py	—	One exact query per (drug, MedDRA term). A count aggregation truncates at openFDA's 100-bucket cap and would silently drop rare pairs. A pre-flight check validates every term string against the database and drops any it does not know, rather than letting it manufacture false zeros.
extract_labels.py	—	Parses SPL XML; records the matching sentence verbatim with its LOINC section. A silent label yields an explicit measured_null row, not no row.
parse_inn_sequences.py	—	Parses WHO INN chemical names, which spell out every residue, into sequence and per-position chemistry.
build_modifications.py	256	One row per nucleotide position: sugar, base, 5-methylation and 3' linkage.
build_literature.py / build_nonclinical.py	—	Curated rows; each carries its verbatim quote.

5 Purification and characterisation of oligo identity

The Challenge asks specifically for the methods used to purify and characterise oligo identity. Because the compounds were made by their sponsors and not here, what can be reported is what each source states — and the honest answer is almost nothing. A full-text sweep of all sixteen committed US prescribing information documents for purity, purification, chromatography, mass-spectrometry, identity and characterisation language returns **no statement about the drug substance in any of them**; every hit is a patient baseline characteristic or an efficacy assay. purity_pct is NOT_REPORTED for all 53 compounds. The two research-reagent sources name a supplier but no method. No purity value has been estimated, inferred from a synthesis platform, or carried across from another compound. Our sibling OligoTox-CNS release reports the same for all 1,839 of its oligonucleotides: this is a property of the published literature, not of the curation.

Identity as recorded here means the sequence together with its per-position chemistry, and both are parsed rather than typed. Sequences come from the WHO INN Recommended lists, whose entries spell out each residue's sugar, base, 5-methylation and 3' linkage longhand. The parse is validated against evidence from outside the INN list and the script exits rather than emit a disagreeing sequence: nusinersen parses to 18 residues with 17 phosphorothioate linkages against a label formula of P17 S17; tofersen to 20 residues with 15 phosphorothioate and 4 phosphodiester linkages, exactly as its label states in words; and tofersen's INN-derived sugar map is identical, position for position, to the map derived independently from the label's own motif sentence.

6 Experimental methods of the source studies

- **Registry adverse-event tables.** Modules declare a frequencyThreshold governing the other-events table, copied into every affected row's ascertainment_basis. 42 CFR 11.48(a)(4)(ii)(A) requires the serious-event table to list *all* serious events with no threshold, which is what makes a trial-level absence a reported zero. **No trial in this release performed protocol-specified ventricular imaging** except the two noted in the narrative.
- **FAERS.** Voluntary spontaneous reports, unvalidated, with no exposure denominator and no causality assessment.
- **Labels.** Postmarketing sections describe reports from a population of uncertain size for which causality cannot always be established.
- **Index cases.** Serial MRI, serial CSF sampling, and a lumbar infusion study measuring resistance to CSF outflow (tominersen); serial MRI, opening pressures and a reduced-dose rechallenge (valeriasen).
- **Disease baseline.** Retrospective matched-cohort study in a ~100-million-person EHR database, 2007–2016, ascertained by ICD-9/10 code.

7 Harmonisation, grading and quality control

hydroceph_grade is a new ordinal column with its own written rubric, not a reuse of an organ-specific scale from another endpoint. Every graded row carries the exact rule in grade_basis. Grade 0 is permitted **only** where ascertainment is measured_null, so "measured and null" is never conflated with "nobody looked". qc/validate.py runs **53 checks** and exits non-zero on any failure, covering primary keys, both foreign keys, controlled vocabularies on twelve columns, the grade-0 rule, numerator ≤ denominator, contiguous nucleotide positions, and byte-identical regeneration of the derived views. Three are genuine cross-source consistency tests rather than format checks: the label's phosphorus count against the stated length, an siRNA duplex's guide strand against the reverse complement of its sense strand, and the modification table's bases against the stored sequence. The data dictionary lives in code and the suite asserts in both directions that every column is documented and every documented column exists — a check added because an earlier revision of our own schema declared three purity fields the builder never emitted.

8 Principal limitations

- **No in vitro rows.** 1,351 human and 10 animal rows, none in vitro. The dataset cannot support in vitro-to-animal extrapolation.
- **Sequences for 13 of 53 compounds.** Duplex siRNAs and morpholinos are refused by the INN parser rather than guessed; one compound's sequence exists only as an image whose bold/underline chemistry encoding does not survive text extraction.
- **All grades are provisional.** No subject-matter expert has reviewed the rubric or its application.
- **100 verified sources retrieved but not extracted**, listed with their retrieval routes in notes/source_backlog.md.